I. PROGRAM OVERVIEW

The Big Data & Machine Learning program at Duy Tan University (DTU) is managed by the School of Computer Science (SCS). It is a flagship academic program designed to prepare students for the Industry 4.0 era, where data serves as the critical engine for economic and technological growth.

1.1. Educational Philosophy

The curriculum employs the CDIO (Conceive - Design - Implement - Operate) model. This ensures that students move beyond theoretical mathematics to develop practical, high-impact technological products. The High-Priority (HP) Talent Program is specifically designed to optimize individual potential through small class sizes and direct mentorship from industry experts.

1.2. Learning Environment

- Modern Laboratories: Equipped with high-performance computing systems and GPUs (NVIDIA RTX/A-series) for training complex Deep Learning models.
- Infrastructure: Students have access to Big Data Clusters (Hadoop/Spark) to practice processing and analyzing massive datasets in real-time.

II. ADMISSION & ENTRANCE REQUIREMENTS (NEW)

To maintain the elite quality of the HP program, the admission process focuses on logical aptitude and academic excellence.

2.1. Admission Subject Combinations

Students can apply using the following standardized groups:

- A00: Mathematics, Physics, Chemistry.
- A01: Mathematics, Physics, English.
- D01: Mathematics, Literature, English.
- D07: Mathematics, Chemistry, English.

I	TRƯỜNG KHOA HỌC MÁY TÍNH & TRÍ TUỆ NHÂN TẠO (TOP 201-250 THẾ GIỚI THEO QS RANKINGS 2025) - https://sca.duytan.edu.vn					
1	7480103	Ngành Kỹ thuật Phần mềm có các chuyên ngành:	*Công nghệ Phần mềm* (Đạt kiểm định ABET)	102	A00 A01 D01 C01 C02 X26	Toán, Lý, Hóa Toán, Lý, Anh Văn, Toán, Anh Văn, Toán, Lý Văn, Toán, Hóa Toán, Anh, Tin
			Thiết kế Games và Multimedia	122		
			Công nghệ Phần mềm Ô tô Thông minh	131		
2	7480202	Ngành An toàn Thông tin có chuyên ngành:	*An toàn Thông tin*	124		
3	7480101	Ngành Khoa học Máy tính có các chuyên ngành:	*Kỹ thuật Máy tính*	128		
			Khoa học Máy tính	130		
4	7480107	Ngành Trí tuệ Nhân tạo có chuyên ngành:	*Trí tuệ Nhân tạo (HP)*	121(HP)		
5	7460108	Ngành Khoa học Dữ liệu có chuyên ngành:	*Big Data & Machine Learning (HP)*	115(HP)		
6	7480102	Ngành Mạng Máy tính & Truyền thông Dữ liệu có các chuyên ngành: (Đạt kiểm định ABET)				
			Kỹ thuật Mạng	101		
			Mạng Máy tính & Truyền thông Dữ liệu	140		

2.2. Enrollment Standards

- Direct Admission: National/International prize winners in STEM subjects.

- Priority: Students with an IELTS score of 5.5+ or high scores in the National High School Exam (usually top 10% nationwide).

III. ACADEMIC ROADMAP (4-YEAR PLAN)

The program spans 4 years (8 semesters), structured into three critical stages:

Stage 1: Foundational Knowledge (Year 1 & 2)

Focuses on logical thinking, core mathematics, and fundamental programming.

- Mathematics for Data Science: Calculus, Linear Algebra, and Probability & Statistics.
- Programming Mastery: Intensive training in Python and C++.
- Data Structures & Algorithms: Optimization for handling millions of records.

Stage 2: Core Technologies (Year 3)

Focuses on distributed systems and advanced machine learning models.

- Big Data Ecosystem: Hands-on experience with Hadoop (HDFS, MapReduce) and Apache Spark.
- Machine Learning (ML): Covers Supervised (Regression, Classification) and Unsupervised learning (Clustering).
- NoSQL Databases: Managing unstructured data with MongoDB and Cassandra.

Stage 3: Specialization & Professional Practice (Year 4)

- Deep Learning (DL): Researching artificial neural networks, NLP, and Computer Vision.
- Internship (OJT): 3-4 months at partners like FPT Software, Enclave, and LogiGear.
- Capstone Project: Solving a real-world data problem (e.g., RAG systems or financial models).

IV. CORE SUBJECTS AND TECHNICAL STACK

Subject	Key Focus	Tools/Languages
Data Mining	Pattern discovery & knowledge extraction.	Weka, Scikit-learn
Cloud Computing	AI system deployment on cloud.	AWS, Google Cloud, Docker
NLP	Chatbots, Sentiment Analysis, RAG.	BERT, Transformers, LangChain
Computer Vision	Face recognition, Autonomous perception.	OpenCV, YOLO, PyTorch
Visualization	Transforming raw data into insights.	PowerBI, Tableau, Seaborn

V. EXPANDED CAREER PATHWAYS (DETAILED)

Graduates are prepared for high-demand roles with starting salaries ranging from $800 to $2,500+ depending on specialization.

1. Machine Learning Engineer: Designing and deploying AI algorithms. Focus on model efficiency and MLOps.
2. Data Scientist: Analyzing complex datasets to drive business strategies using statistical inference.
3. Big Data Engineer: Architecting large-scale data pipelines and managing Data Lakes.
4. AI Research Scientist: Developing new neural network architectures and publishing academic papers.
5. NLP Specialist: Developing Large Language Models (LLMs) and Vector Databases.
6. CV Specialist: Working on self-driving technology and medical imaging diagnostics.